

SHREEVIKAS JAGADISH

AI Engineer | Machine Learning | Predictive Modeling | Decision Intelligence

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SUMMARY

Artificial Intelligence Engineer with over 4 years of experience architecting scalable Machine Learning, Computer Vision, and Generative AI systems for enterprise manufacturing. Expert in leveraging Python, PyTorch, AWS, and Azure ML to deploy high-throughput predictive models, dynamic RAG workflows, and robust MLOps infrastructure (Docker, Kubernetes) that automate complex data pipelines, drive multimillion-dollar cost efficiencies, and elevate product reliability.

PROFESSIONAL WORK EXPERIENCE

Procter & Gamble, United States

January 2026 – Present

Artificial Intelligence Engineer

- Architected high-speed computer vision inspection pipelines using Python, OpenCV, and PyTorch, leveraging CUDA-accelerated GPU clusters on AWS SageMaker to train defect-detection models that reduced product recall rates by 26%.
- Developed and shipped an Agentic AI supply chain assistant utilizing LangGraph, OpenAI SDK, and LangSmith for observability, orchestrating multi-agent workflows that resolved procurement bottlenecks and slashed manual intervention by 30%.
- Constructed enterprise Retrieval-Augmented Generation (RAG) systems using LangChain, LlamaIndex, and FAISS vector indices, automating cross-departmental policy extraction to decrease internal HR and IT support resolution times by 40%.
- Fine-tuned open-source Large Language Models leveraging QLoRA, PEFT, and Hugging Face Transformers, automating unstructured vendor contract summaries and cutting manual compliance auditing efforts by 35%.
- Orchestrated scalable data pipeline feature stores on Databricks, Delta Lake, and Apache Airflow, uncoupling legacy data silos to accelerate downstream AI model training iterations across global supply chain squads.
- Built production-ready RESTful inference microservices using FastAPI, Docker, and Kubernetes on AWS, enabling high-concurrency model execution to save over \$1.2M in annual cloud infrastructure overhead.

Bosch, India

March 2021 – July 2024

Machine Learning Engineer

- Constructed high-accuracy time-series forecasting models using XGBoost, scikit-learn, and AWS S3, predicting industrial spare parts demand surges to decrease inventory holding costs by 22%.
- Designed document processing OCR pipelines incorporating PaddleOCR, PyMuPDF, and OpenCV, automating field-level extraction from complex engineering blueprints to cut manual review cycles by 45%.
- Formulated automated anomaly detection algorithms utilizing Isolation Forest and PyTorch, monitoring telemetry data across connected automotive systems to elevate overall product reliability and customer safety.
- Implemented continuous integration and deployment (CI/CD) MLOps pipelines using MLflow, Airflow, and Azure ML, standardizing model monitoring and drift detection to compress retraining turnarounds by 3 weeks.
- Developed scalable NLP text classification microservices using BERT models, Python, and PostgreSQL, automatically categorizing multi-lingual technical warranty claims to boost root-cause analysis speed by 30%.
- Optimized heavy SQL queries and distributed ETL data pipelines on Apache Spark and Snowflake, improving feature engineering throughput to support high-frequency predictive model execution.
- Executed A/B testing frameworks and statistical hypothesis testing methodologies using Python and Tableau, validating algorithmic calibration updates to capture an immediate 14% increase in manufacturing yield efficiency.

EDUCATION

Master of Science in Information Technology & Management | Illinois Institute of Technology, USA

May 2026

Bachelor in Computer Science | Visvesvaraya Technological University, India

August 2023

SKILLS

Programming & Scripting: Python, Java, SQL, Bash, PowerShell.

Deep Learning & Generative AI: LLMs, CNN, NLP, Computer Vision, GPT-4o, Gemini API, Agentic AI (LangGraph, CrewAI), Fine-Tuning (PEFT, LoRA, QLoRA), Prompt Engineering, OpenAI SDK, Function/Tool Calling, Whisper ASR.

RAG & Retrieval: LlamaIndex, LangChain, Embeddings (Hugging Face, all-MiniLM-L6-v2), Vector Databases (ChromaDB, FAISS), Chunking Strategies, Reranking, Retrieval Evaluation.

Machine Learning: scikit-learn, XGBoost, Logistic Regression, Isolation Forest, Anomaly Detection, K-Mean Clustering, Statistical Modeling, A/B Testing, TensorFlow, PyTorch, Time Series Forecasting, Feature Engineering, AUC-ROC, F1 Score.

Document AI & OCR: PyMuPDF, PaddleOCR, Tesseract, EasyOCR, OpenCV, PDF Parsing, Regex Field Extraction.

Data Engineering: Databricks, Snowflake, Delta Lake, Apache Spark, Kafka, dbt, Feature Engineering, ETL/ELT Pipelines.

Cloud, MLOps & DevOps: MLflow, FastAPI, REST APIs, Apache Airflow, Docker, Kubernetes, AWS (SageMaker, S3, EC2, Lambda), Azure (ML, Data Factory), GCP, CUDA (GPU Acceleration), LangSmith (Observability), CI/CD, Terraform, Git.

Databases: PostgreSQL, MS SQL Server, Oracle, MySQL, MongoDB (NoSQL), Amazon Redshift.

BI, Visualization & Tools: Tableau, Power BI, Plotly, Seaborn, Matplotlib, Alteryx, ServiceNow, Excel, Agile/Scrum, Jira.

PROJECTS

Autonomous Industrial Quality Assurance & Defect Intelligence System

- Tasked with resolving high manual inspection costs and delayed defect detection on high-speed automated assembly lines that caused unexpected product batch rejections.
- Built a real-time computer vision and deep learning inspection pipeline using PyTorch, OpenCV, Docker, and AWS SageMaker, processing 500+ frames per second to reduce packaging defect leakage by 32% and save \$850K in scrap material expenses.

Manufacturing Process Quality Intelligence System | Python, Scikit Learn, SQL, Pandas, PostgreSQL, Tableau

- Required to streamline manual data entry from complex, layout-heavy legal and vendor contracts across fragmented enterprise storage repositories.
- Engineered an end-to-end Document AI platform using PaddleOCR, PyMuPDF, LangChain, and ChromaDB vector search, enabling semantic document Q&A and field parsing to accelerate contract auditing workflows by 40%.